



# Progress on Defect Predictor

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# Implementation

- Document code
- Update README.md
- Fixed calculate\_minor\_contributor()
- Test with harness and ground truth
- Implemented weight optimizing function

# Revisit Factors in the Formula

- Recency (Date committed): 22.5%
- Frequency (How often the files are changed): 30%
- Number of lines modified: 15%
  - Lines added + lines deleted
- How many lines of code (loc) in the file: 10%
- Cyclomatic Complexity: 12.5%
- Number of minor contributors: 10%

# Optimization harness

- Ground truth having more files (top 5 instead of top 3)
  - Helps with weight optimizing
  - Closer to actual accuracy (Right now accuracy is only 0%, 20%, 50%, 100%)
- Parametrize weights (Have one parameter at 25% and everything else 15%)
- Loop through different weight sets
- Loop through different config.jsons
- Call Pythonminer for each config.json
- Calculate risk scores -> accuracies
- Store accuracies for each sets of weights across all config.jsons
- Calculate accuracy average
- Return the set of weights with the highest average accuracy



# Harness bug?

```
Calculating risk scores...
```

```
Risk Scores (File Name -> Risk Value):
conf.py -> 0.03860078428602
fourier_continuation.py -> 0.026152051951956738
model_dispatcher.py -> 0.1867303455479052
torch_setup.py -> 0.04316513333075183
train_darcy.py -> 0.23443813979077685
skip_connections.py -> 0.06840246560519658
fno_block.py -> 0.6417926449315449
tfno.py -> 0.49716840536512663
__init__.py -> 0.054806914647385525
uno.py -> 0.48657806599244746
padding.py -> 0.37463271483223043
resample.py -> 0.28752139213923855
spectral_convolution.py -> 0.4382953711352021
train_navier_stokes.py -> 0.26762724169004376
__init__.py -> 0.10684840368179856
normalization_layers.py -> 0.14622872515788762
plot_SFNO_swe.py -> 0.2927244515416826
spherical_swe.py -> 0.1918896227361148
plot_UNO_darcy.py -> 0.17995928235405745
spherical_convolution.py -> 0.3795399374263474
fno.py -> 0.56616522993168
padding.py -> 0.29920907289957116
```

```
Top 3 Risky Files:
```

```
- fno_block.py -> 0.6417926449315449
- fno.py -> 0.56616522993168
- tfno.py -> 0.49716840536512663
```

```
Comparing Top 3 Risky Files with Ground Truth...
```

```
False Positives (Files predicted but not in ground truth):
tfno.py
```

```
False Negatives (Files in ground truth but not predicted):
spectral_convolution.py
trainer.py
local_no_block.py
uno.py
```

```
Accuracy: 0.29
```

We believed the harness is grabbing the ground truth of the repos with the same name above it instead of taking into account the specific data range.

# What's next?

- Wrapping Up
  - Fine tune a bit more based on parameter testing for weight
  - Local storage of feature scores (so pydriller only has to run 4 times instead of 28)
  - Wrap up documentation for code and README.md
  - Start one page report